Manual: Bug Report Classification Tool

Overview

This tool classifies GitHub bug reports as performance-bug-related (1) or not (0). It trains and evaluates two approaches side by side:

- Baseline: Naive Bayes + TF-IDF
- Proposed: LinearSVC + TF-IDF with class balancing

Results, figures, and raw scores are saved automatically to an output folder of your choice.

Setup

1. Install Python 3.8 or higher.
2. Install the required packages:

```
pip install numpy pandas scikit-learn scipy matplotlib
```

3. Download the 5 CSV files from the official lab repository or my github repository:

   https://github.com/ideas-labo/ISE/tree/main/lab1 or
   https://github.com/aasfimanwar/ise-coursework (data folder)
   Note: I have changed the file name from incubator-mxnet.csv to mxnet.csv in my repository

4. Place all 5 CSV files in a folder named data/ next to classify.py

Running the Tool

```
python classify.py --data_dir ./data --out_dir ./results
```

Output Files

After running, the results/ folder will contain:

- boxplot_f1.pdf
- boxplot_precision.pdf
- boxplot_recall.pdf
- mean_f1_bar.pdf
- raw_results.json
- results_summary.csv

Interpreting Results

- Precision: of all reports predicted as performance bugs, how many actually are?
- Recall: of all actual performance bug reports, how many did the tool find?
- F1: harmonic mean of precision and recall the primary comparison metric.
- p-value < 0.05: the difference between baseline and proposed is statistically significant.
- VD-A > 0.56 = small effect; > 0.64 = medium; > 0.71 = large.